

Replication Guide for “Effects of Subsidies on Welfare and Market Structure in the US Broadband Industry”

Sidi Mohamed Sawadogo

June 2026

1 Purpose

This document provides a concise replication guide for the 2020 broadband demand, fixed-cost, and counterfactual simulations. It is intended as a run-order guide for a replication package rather than as a full methodological appendix. The main pipeline estimates nested-logit demand and marginal costs, constructs Fan–Yang profit bounds, estimates a seven-parameter fixed-cost model, draws 50 fixed-cost vectors from the confidence region, and simulates baseline, ACP, and BEAD portfolio best-response outcomes.

The core output of the replication package is a set of draw-level and trimmed intervals for welfare, entry, prices, quantities, and concentration. All counterfactual simulations use the same fixed-cost draw index, so draw-level comparisons can be computed by matching on **draw**.

2 Folder Structure

The working folder is:

```
/Users/sidimohamedsawadogo/Package thesis/JMP/Code
```

The intended replication structure is:

Folder	Contents
data/	Input data, intermediate CSV files, final CSV tables, and run logs.
graphic/	PDF and PNG figures.
documentation/	TeX and PDF documentation, including this guide.
Archives/	Archived or alternative scripts not needed for the main run.

Scripts should be run from the **Code** folder. Some older scripts may still write selected intermediate files to the root of **Code**; when archiving the replication package, store CSV outputs in **data/**, figures in **graphic/**, and TeX/PDF documentation in **documentation/**.

3 Execution Order

The main replication sequence is:

1. 02. `nested_logit_demand_supply_2020.py`
2. 03. `estimate_profit_bounds_fan_yang_2020.py`
3. 04. `compute_fixed_cost_fan_yang_2020.py`

4. 06. `simulate_baseline_portfolio_br_outcomes_2020.py`
5. 07. `simulate_acp_portfolio_br_nash_counterfactual_2020.py`
6. 08. `simulate_bead_portfolio_br_nash_counterfactual_2020.py`
7. 09. `cost_benefit_analysis_2020.py`

Recommended smoke-test commands (run from the Code folder):

```
python3 "03. estimate_profit_bounds_fan_yang_2020.py" --max-tracts 10 --workers 1
python3 "04. compute_fixed_cost_fan_yang_2020.py" --n-tracts 500 --output-root /tmp/fy_test
python3 "06. simulate_baseline_portfolio_br_outcomes_2020.py" --max-draws 2 --max-markets 100
python3 "07. simulate_acp_portfolio_br_nash_counterfactual_2020.py" --max-draws 2 --max-markets 100
python3 "08. simulate_bead_portfolio_br_nash_counterfactual_2020.py" --max-draws 2 --max-markets 100
```

Script 04 smoke tests should use `-output-root` to avoid overwriting full-sample outputs. Scripts 06–08 smoke tests write to the real `data/` folder; use only for quick syntax/logic checks.

The full Fan–Yang profit-bound step (scripts 03–04) can take 1–2 hours on 70 854 tracts; scripts 06–08 take 30–90 minutes each with 50 draws.

4 Script-by-Script Summary

Script	Main task	Key outputs
02	Estimates nested-logit demand and supply primitives.	<code>nested_logit_parameters_2020.csv</code> , <code>nested_logit_markup_2020.csv</code> , <code>nested_logit_mc_ols_nofe_2020.csv</code> .
03	Computes Fan–Yang expected variable-profit bounds using empirical profit-shock draws.	<code>profit_bounds_fan_yang_2020.csv</code> , <code>potential_products_fan_yang_2020.csv</code> .
04	Estimates seven fixed-cost parameters via moment inequalities; draws 50 FC vectors from the Andrews–Soares confidence region; writes comprehensive documentation PDF.	<code>fixed_cost_simulation_draws_2020.csv</code> , <code>fixed_cost_fan_yang_2020_paper_table.csv</code> , <code>fixed_cost_fan_yang_2020_documentation.pdf</code> .
06	Solves baseline portfolio best-response equilibria for each draw; reports welfare, prices, HHI, level counts (products H/L, active firms, coverage, averages per tract), and market-share metrics.	<code>baseline_portfolio_br_nash_draw_intervals_2020.csv</code> , <code>baseline_portfolio_br_nash_trimmed_bounds_2020.csv</code> , <code>baseline_portfolio_br_nash_documentation.pdf</code> .
07	Simulates ACP consumer-price discounts (\$10/\$20/\$30); same outputs as script 06 plus % change bounds vs baseline; loads FPL eligibility from ACS file.	<code>acp_portfolio_br_nash_draw_intervals_2020.csv</code> , <code>acp_portfolio_br_nash_documentation.pdf</code> .
08	Simulates BEAD as a proportional fixed-cost reduction (25/50/75%); same outputs as script 07.	<code>bead_portfolio_br_nash_draw_intervals_2020.csv</code> , <code>bead_portfolio_br_nash_documentation.pdf</code> .
09	Combines all draw-level results into CBA tables (NPV, BCR, net surplus) and figures; writes comprehensive documentation PDF.	<code>cba_draw_level_2020.csv</code> , <code>cba_trimmed_bounds_2020.csv</code> , <code>cba_doc.pdf</code> .

5 Data Sources

Table 1: Primary data sources

Dataset	Provider	Vintage	Link / Access
FCC Broadband Deployment Data (Form 477)	Federal Communications Commission	December 2020	https://www.fcc.gov/general/broadband-deployment-data-fcc-form-477
FCC Urban Rate Survey (URS)	Federal Communications Commission	2020	https://www.fcc.gov/economics-analytics/industry-analysis-division/urban-rate-survey-data-resources
FCC Consumer Survey on Services (CSS) / Broadband Subscription Shares	Federal Communications Commission	2020	https://www.fcc.gov/economics-analytics/industry-analysis-division/broadband-subscription-data
ACS 5-year Estimates — Demographic & Economic Characteristics	U.S. Census Bureau	2019 (5-yr)	https://data.census.gov (tables B01003, B19013, B25003, B15003, B01001)
ACS Table C17002 — Ratio of Income to Poverty Level	U.S. Census Bureau	2019 (5-yr)	https://data.census.gov/table/ACSDT5Y2019.C17002
ACS Table B28002 — Broadband / Internet Access	U.S. Census Bureau	2019 (5-yr)	https://data.census.gov/table/ACSDT5Y2019.B28002
Census TIGER/Line Shapefiles — Census Tracts	U.S. Census Bureau	2020	https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html

Notes on access. FCC Form 477 data can be downloaded from the FCC broadband map portal or the bulk download page. The URS workbook is distributed as an Excel file (.xlsx); if the system Python lacks Excel support, convert to CSV first. ACS tables can be retrieved via the Census API at <https://api.census.gov/data> with geography `tract` and year 2019.

6 Required Inputs

The main upstream input for the demand stage is `data/finaldataset.csv`. Script 02 creates the preferred nested-logit outputs used by later scripts. The minimum downstream input files are:

- `data/nested_logit_parameters_2020.csv`: preferred demand parameter file with `alpha_price_100` and `rho`.
- `data/nested_logit_markup_2020.csv`: product-level demand and cost residuals used for Fan–Yang profit-shock draws.
- `data/nested_logit_mc_ols_nofe_2020.csv`: marginal-cost regression coefficients used for missing-cost imputation.
- `profit_bounds_fan_yang_2020.csv`: expected profit bounds used in fixed-cost estimation.
- `fixed_cost_simulation_draws_2020.csv`: 50 fixed-cost columns, `fc_draw_001` through `fc_draw_050`.
- `data/potential_products_fan_yang_2020.csv`: product primitives used by the baseline and policy simulations.

Script 07 (ACP) requires `share_pop_below_200_fpl` per tract. If this column is absent from the draw file, it is loaded automatically from `data/acs_tract_2020_fpl.csv` (ACS 2019 Table C17002). Missing values are

imputed by county median; any remaining gaps use the overall sample median.

7 Key Modeling Choices

Demand and Profit Bounds

Demand is estimated with a nested-logit model. The downstream simulations read the preferred specification from `data/nested_logit_parameters_2020.csv` rather than using hard-coded demand parameters. The preferred parameter values in the current package are approximately

$$\hat{\alpha}_{100} = -1.4907, \quad \hat{\rho} = 0.8629.$$

Fan–Yang profit bounds are expected variable-profit bounds. They are not deterministic zero-shock profits: the run log records 100 empirical profit draws using paired demand and cost residuals. The relevant log is `data/fan_yang_run_log_2020.txt`.

Fixed-Cost Model

The fixed-cost model has seven parameters:

$$(\theta_S, \theta_M, \theta_L, \theta_H, \sigma_S, \sigma_M, \sigma_L).$$

The three θ parameters vary by market-size tertile, θ_H is the high-speed tier premium, and the three σ parameters are market-size-specific fixed-cost shock standard deviations.

Important settings in script 04 (changed from earlier versions):

- $\sigma_b \geq 0$ only — no lower-bound constraint other than strict positivity via the exp transform. The optimizer and stochastic search are both unconstrained on $\log \sigma_b$.
- **Confidence-set search:** 50 local chains $\times$ 50 steps on all available CPUs, followed by 100 expansion chains seeded from boundary points. Chains start from a grid of diverse σ values $\{10^{-10}, 100, 500, 1000, 5000, 10000\}$ combined with the optimizer’s θ estimates, cycling across chains. This prevents degenerate starts when the optimizer converges to $\sigma \approx 0$.
- **Perturbation scale:** $\max(250, 0.2|\hat{\theta}_b|)$ for each θ parameter; 0.2 (log scale) for each σ . Expansion phase uses $5\times$ these scales.
- **Seeds:** optimizer starts use 7 starting vectors ($\sigma \in \{250, 500, 1000, 2500, 5000, 10000\}$ plus data-based init); confidence-set seed 20240525; simulation seed 20240524.

The simulation uses 50 accepted parameter vectors from the confidence region and draws fixed costs so that observed portfolios are rationalized as equilibria under the Fan–Yang bounds.

Baseline Portfolio Game

For each fixed-cost draw, the baseline script solves a tract-level portfolio game. Each provider chooses among feasible product portfolios, such as no product, low tier, high tier, or both tiers. Best responses are iterated from several starting configurations. Distinct fixed points that pass a firm-by-firm Nash check define the market-level equilibrium set. When multiple equilibria are found, the script reports lower and upper outcome bounds.

ACP Counterfactual

ACP is implemented as a targeted consumer-price discount:

$$p_{jt}^c(a) = \max\{p_{jt}^g - a, 0.01\}, \quad a \in \{10, 20, 30\}.$$

The market is split into an eligible segment and an ineligible segment:

$$M_t^{elig} = M_t \times \text{share_pop_below_200_fpl}_t, \quad M_t^{inelig} = M_t \times (1 - \text{share_pop_below_200_fpl}_t).$$

Only the eligible segment receives the discount and faces $p_{jt}^c(a)$. The ineligible segment keeps the original consumer price p_{jt}^g . The gross provider price and marginal cost determine the provider margin, fixed costs are unchanged, and government outlay is the discount times eligible quantity.

BEAD Counterfactual

BEAD is modeled as a stylized fixed-cost subsidy:

$$FC_{jt}^{cf}(\tau) = (1 - \tau)FC_{jt}, \quad \tau \in \{0.25, 0.50, 0.75\}.$$

This captures the supply-side deployment incentive of infrastructure grants. It is not a literal simulation of all statutory BEAD targeting rules. The CBA step computes the fiscal cost as

$$G^{BEAD}(\tau) = \frac{\tau}{1 - \tau} \times PFC(\tau) \times 12,$$

where $PFC(\tau)$ is the counterfactual private fixed cost of active products.

8 Main Outcomes

Scripts 06–08 report draw-level lower/upper bounds for the following metrics. Script 09 adds CBA metrics. All bounds use $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ across 50 draws.

Metric group	Variables
Welfare	Consumer surplus, producer surplus (\$B/month)
Prices & costs	Avg price, avg marginal cost (\$/month)
Concentration	Market-size-weighted avg tract HHI (0–10,000)
Market share	Avg product market share $\bar{s}_{jt}$; overall take-up rate
Level counts	Total products offered; H products; L products; active firms (firm-market pairs); covered markets (tracts ≥ 1 product)
Avg per tract	Avg products/tract; avg H/tract; avg L/tract; avg firms/tract
Entry (changes)	New products H/L; new firm-markets; % entry rates vs baseline
ACP only	Government outlay; eligible/ineligible quantities; total surplus net of outlay

Scripts 07 and 08 additionally report a **percentage-change table** vs baseline. For baseline interval $[a, b]$ and counterfactual interval $[c, d]$, the two boundary-by-boundary percentage changes are computed and sorted:

$$[\% \Delta^L, \% \Delta^U] = \left[\min\left(\frac{c - a}{a}, \frac{d - b}{b}\right), \max\left(\frac{c - a}{a}, \frac{d - b}{b}\right) \right] \times 100.$$

Lower boundary a pairs with lower CF boundary c ; upper baseline b pairs with upper CF boundary d . Results are then sorted to ensure $\% \Delta^L \leq \% \Delta^U$.

9 Cost-Benefit Analysis

Script 09 merges each policy result with the baseline result by **draw**. Interval arithmetic is conservative:

$$\underline{\Delta Y}^r = \underline{Y}_{cf}^r - \overline{Y}_{base}^r, \quad \overline{\Delta Y}^r = \overline{Y}_{cf}^r - \underline{Y}_{base}^r.$$

For BEAD, welfare changes are monthly flows converted to annual flows and then discounted over a 25-year horizon. For ACP, government outlays and welfare changes are annual recurring flows, so the main statistics are annual net surplus and benefit-cost ratios.

10 Verification Checklist

Before reporting final results, check the following:

1. Script 03 run log confirms empirical shock draws (not zero-shock).
2. Script 04 writes exactly 50 draw columns in `fixed_cost_simulation_draws_2020.csv`.
3. `fixed_cost_simulation_draws_2020_parameters.csv` reports `all_rows_best_response = True` for all 50 draws.
4. Script 04 confidence-set search yields ≥ 50 accepted points (check printed output: “Total accepted: N”).
5. Scripts 06–08 all read the same `nested_logit_parameters_2020.csv` and the same `fixed_cost_simulation_draws_2020.csv`.
6. Draw-level outputs from scripts 06–08 contain the new level-count columns: `offered_products_H_lower`, `active_firms_lower`, `avg_products_per_tract_lower`, etc. (required for % change table and CBA).
7. Script 07 prints no `ValueError` about `share_pop_below_200_fpl`; check that FPL imputation ran (county-median fallback message printed if any tract was missing).
8. Script 09 reads from `data/` and writes outputs to `data/` and `documentation/`. Verify `data/cba_draw_level_2020.csv` exists after run.
9. All documentation PDFs recompile without errors (check `documentation/*.log` for LaTeX errors).

11 Reproducibility Notes

Runtime. Script 04 takes roughly 1–2 hours on 70 854 tracts (parameter estimation + confidence-set search). Scripts 06–08 take 30–90 minutes each with 50 draws and 10 workers.

Draw consistency. Do not mix outputs from different fixed-cost draw files. The CBA merges on `draw`; `draw` `r` must correspond to the same fixed-cost realization in scripts 06, 07, 08. When re-estimating fixed costs or changing the 50-draw selection, rerun scripts 06–09 in order.

Output files. Every script now writes its documentation as a compiled PDF in `documentation/` and figures in `graphic/`. Key documentation files:

- `fixed_cost_fan_yang_2020_documentation.pdf` — full replication details for script 04 (seeds, perturbation scales, weight-function cutoffs, sigma constraints).
- `baseline/acp/bead_portfolio_br_nash_documentation.pdf` — per-draw tables, descriptive statistics, equilibrium diagnostics, and figures for scripts 06–08.
- `cba_doc.pdf` — full cost-benefit analysis (script 09).

External data dependency. Script 07 reads `data/acs_tract_2020_fpl.csv` for eligible-population shares. This file must be present; it is *not* generated by the pipeline.

Python version. The pipeline requires Python 3.9+. Tested on Python 3.14. Avoid Python 3.7 or earlier (`from __future__ import annotations` is required for dataclass type hints).